

Improving four-class *Fermi*-LAT source classification through spectral feature engineering and covariate shift quantification

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Abstract

We present a machine learning classification study of the 2,423 unassociated gamma-ray sources in the *Fermi*-LAT 14-year Fourth Source Catalogue (4FGL-DR4; Ballet et al. 2023), discriminating between four source classes: BL Lacertae objects (BLL), Flat Spectrum Radio Quasars (FSRQ), young pulsars (PSR), and millisecond pulsars (MSP). Building on the seven-feature baseline of Malyshev & Bhat (2023), we introduce three modifications that together raise the 10-fold cross-validated balanced accuracy of a Random Forest classifier from $84.5 \pm 3.3\%$ to $86.5 \pm 3.4\%$ on the same labeled sample of 2,630 sources (Wilcoxon signed-rank test: $p = 0.029$). The core modification is the replacement of the naive power-law index `PL_Index` with the log-parabola spectral index evaluated at 1 GeV, $\Gamma_{1\text{ GeV}} \equiv \Gamma_{\text{LP}} + 2\beta_{\text{LP}} \ln(1\text{ GeV}/E_{\text{piv}})$ (Malyshev & Bhat, 2023),

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supplemented by the log-parabola peak energy $\log E_{\text{peak}}^{\text{LP}}$ and the fractional variability F_{var} . The gain is concentrated in the FSRQ class ($\Delta F_1 = +0.026$), whose separation from BLL improves when spectral curvature is measured at a physically consistent reference energy rather than at the individually fitted pivot energy; a direct $\Gamma_{1\text{ GeV}}\text{--}\log E_{\text{peak}}^{\text{LP}}$ projection confirms that this gain traces the classical blazar sequence. We stress-test the feature-set comparison with repeated cross-validation (5 random seeds $\times$ 10 folds, 50 paired estimates) and independent 70/30 hold-out resampling (15 repeats), both of which confirm the gain at higher significance ($p = 8.8 \times 10^{-4}$ and $p = 0.022$, respectively) than the single-split test alone. Permutation importance corroborates the Gini-based feature ranking, with the log-parabola curvature β_{LP} and the spectral index $\Gamma_{1\text{ GeV}}$ remaining the two most informative features under both metrics. We quantify the covariate shift between labeled and unassociated sources across all ten features using Wasserstein distances and KS tests, and find that a binary shift classifier achieves $\text{AUC} = 0.985$, confirming that the two populations are nearly perfectly separable in feature space. Of the 2,423 unassociated sources, 34.0% fall below the 5th percentile of the training significance distribution and are designated out-of-distribution (OOD); their classifications carry reduced reliability and are clearly flagged in our published table. An out-of-fold calibration check shows that our adopted high-confidence threshold, $P_{\text{max}} \geq 0.80$, corresponds to an observed accuracy of 96.1% among in-distribution sources, validating its use for follow-up prioritisation. Applying the 10-feature Random Forest (retrained deterministically; see Data Availability) to the 1,598 remaining in-distribution sources, we identify 541 high-confidence candidates ($P_{\text{max}} \geq 0.80$): 274 BLL, 61 FSRQ, 74 PSR, and 132 MSP. All posterior probability tables, the trained model, and the analysis and validation code are provided as supplementary machine-readable data.

Key words: gamma-ray astronomy — catalogues — methods: statistical — BL Lacertae objects: general — pulsars: general — quasars: general — surveys

1 Introduction

The *Fermi*-LAT Large Area Telescope (LAT) has operated continuously since 2008 (Atwood et al., 2009), surveying the entire sky above 50 MeV. Its source catalogues have been updated through a succession of releases, from the Second *Fermi*-LAT LAT Catalog (Nolan et al., 2012) through the Third Catalog (Acero et al., 2015), the Fourth Catalog (Abdollahi et al., 2020) and its associated catalogue of active galactic nuclei (Ackermann et al., 2015; Ajello et al., 2020). The most recent source list, 4FGL-DR4 (Ball et al., 2023), contains 7,195 gamma-ray sources. Among these, roughly 30 per cent – 2,423 sources – carry no association with any catalogued counterpart at other wavelengths. Identifying the likely source classes among these unassociated detections advances both population statistics and the scheduling of radio and X-ray follow-up campaigns, such as the Swift-XRT/UVOT multiwavelength surveys that have independently characterised counterparts to hundreds of unassociated sources (Kerby et al., 2021).

The dominant identified classes are extragalactic: 1,490 BL Lacertae objects (BLL), 820 Flat Spectrum Radio Quasars (FSRQ), and 1,623 blazar candidates of uncertain type (BCU). The second-most important identified population is Galactic: 141 young pulsars (PSR) and 179 millisecond pulsars (MSP), as catalogued in the Third *Fermi*-LAT Pulsar Catalog (Smith et al., 2023), building on the earlier Second *Fermi*-LAT Pulsar Catalog (Abdo et al., 2013) and the substantial MSP discoveries enabled by radio timing campaigns targeting *Fermi*-LAT unassociated sources (Guillemot et al., 2016). Because the four confirmed classes – BLL, FSRQ, PSR, MSP – differ profoundly in their underlying physics and their multi-wavelength counterpart types, a reliable four-class classification is considerably more informative for follow-up work than the binary AGN/non-AGN separation that has dominated the literature (Ackermann et al., 2012; Mirabal et al., 2012; Doert & Errando, 2014).

Machine learning classification of *Fermi*-LAT-LAT sources has been studied extensively. Chiaro et al. (2016) applied artificial neural networks to separate BLL from FSRQ in 3FGL, an approach later refined by Lefaucheur & Pita (2017) using multivariate boosted-tree and

neural-network classifiers on 3FGL blazar candidates. Kang et al. (2019) and Kováčević et al. (2020) obtained accuracies of 91–93 % for binary blazar classification on 4FGL. Luo et al. (2020) systematically investigated how feature selection and sample size affect classification performance, motivating the compact feature sets used in later work, while Finke et al. (2021) explored classification directly from unbinned energy and time spectra with deep learning. Zhu et al. (2021) and Coronado-Blázquez (2022) extended multi-class and gradient-boosted classification to the full 4FGL/4FGL-DR3 unassociated population, and Bhat & Malyshev (2022) constructed probabilistic multi-method catalogues combining Random Forest, boosted trees, and neural-network posteriors. Cooper et al. (2023) reached 91.2 % using SuperLearner ensembles on 4FGL-DR3, and Zhu et al. (2024) used a multi-method voting strategy on the same release, demonstrating that ensemble approaches improve stability. Bhatta et al. (2024) explored self-supervised pre-training on 4LAC-DR3. Three-class and four-class problems – adding pulsars alongside the two blazar types – were addressed by Saz Parkinson et al. (2016) on 3FGL ($\sim 70\%$ balanced accuracy) and by Zhu et al. (2024) on 4FGL-DR3 ($\sim 72\%$).

Two methodological issues have received increasing attention but remain incompletely resolved. First, the choice of spectral index feature matters more than is commonly appreciated. Malyshev & Bhat (2023) noted that the naive power-law index `PL_Index`, computed for all sources regardless of their preferred spectral model, is a suboptimal discriminator because for sources best described by a log-parabola (LP) or sub-exponential cutoff (PLEC) spectrum, `PL_Index` conflates two physically distinct quantities: the intrinsic low-energy slope and the mean slope artificially flattened by spectral curvature. Their prescription of evaluating the LP spectral index at a common reference energy of 1 GeV, $\Gamma_{1\text{ GeV}}$, partially resolves this issue for blazar-like sources. We show here that supplementing $\Gamma_{1\text{ GeV}}$ with the peak energy of the LP SED, $\log E_{\text{peak}}^{\text{LP}}$, provides additional separation between BLL and FSRQ.

Second, the labeled training set (associated sources) and the classification target (unassociated sources) are not drawn from the same distribution. Malyshev (2023) quantified this covariate shift and its effect on classification performance for the 4FGL sample; we ex-

tend this analysis with an explicit per-feature Wasserstein/KS characterisation and a binary shift-classifier AUC for 4FGL-DR4, following the framework of Shimodaira (2000).

The present paper makes the following specific contributions: (i) a systematic, statistically tested comparison of the Malyshev & Bhat (2023) seven-feature set with a ten-feature extension on the full 4FGL-DR4 labeled sample; (ii) a complete, per-feature covariate shift characterisation for 4FGL-DR4; (iii) an explicit OOD designation for unassociated sources that fall outside the support of the training distribution; and (iv) a probabilistic four-class classification table for all 2,423 unassociated 4FGL-DR4 sources, with per-source confidence scores and OOD flags, made available as a machine-readable catalogue.

We do not claim to supersede Malyshev & Bhat (2023) or Zhu et al. (2024); rather, we extend both studies to 4FGL-DR4 with a rigorously tested feature set and a transparent handling of distribution shift.

The paper is organised as follows. Section 2 describes the data and feature construction. Section 3 presents the covariate shift analysis. Section 4 describes the classifiers and validation strategy. Results are given in Section 5. Section 6 discusses the physical interpretation and limitations. Section 7 summarises our conclusions.

2 Data and feature construction

2.1 The 4FGL-DR4 catalogue

We use the 4FGL-DR4 catalogue (Ball et al., 2023) in its FITS release `gll_psc_v35.fit`, downloaded from the Fermi Science Support Center. The main source table contains 7,195 rows and 79 spectral, positional, variability, and association columns. We read this file with a bespoke pure-Python FITS binary-table parser (Appendix A) in environments where `ASTROPY` is unavailable, verified against the published source statistics.

The labeled sample comprises all sources with a non-blank `CLASS1` field whose stripped lower-case value falls in `{b11, BLL, fsrq, FSRQ, psr, PSR, msp, MSP}`: a total of 2,630

sources (BLL: 1,490; FSRQ: 820; PSR: 141; MSP: 179). Both lower-case (likely association) and upper-case (firm identification) entries are included, since the association probability for lower-case classes in 4FGL-DR4 exceeds 80 per cent by construction (Ball et al., 2023). Blazar candidates of uncertain type (bcu/BCU; 1,623 sources) are treated as a separate prediction-only set. The 2,423 sources with a blank CLASS1 field are the primary classification target.

2.2 Feature set

Our ten features are listed in Table 1 and motivated below.

The first seven features follow Malyshev & Bhat (2023) exactly: $\log F_E$ ($\log_{10}$ energy flux in $\text{erg cm}^{-2} \text{s}^{-1}$), $\log \sigma_{F_E}$ (its log uncertainty), $\log \sqrt{TS}$ (log detection significance), $\Gamma_{1 \text{ GeV}}$ (LP spectral index at 1 GeV; see below), β_{LP} (log-parabola curvature parameter), σ_{curv} (log-parabola curvature significance), and $\log V$ (log variability index). Three features are added by the present work: $|b|$ (absolute Galactic latitude), $\log E_{\text{peak}}^{\text{LP}}$ (log peak energy of the LP SED), and F_{var} (fractional variability amplitude).

The $\Gamma_{1 \text{ GeV}}$ feature. The 4FGL-DR4 catalogue fits three spectral models to each source: a power law (PL), a log-parabola (LP), and a sub-exponential cutoff power law (PLEC). The column `PL_Index` contains the PL best-fit index regardless of which model is preferred. For sources whose preferred model is LP or PLEC, this column conflates the intrinsic low-energy slope with the effect of spectral curvature. For the LP model,

$$\frac{dN}{dE} \propto \left(\frac{E}{E_{\text{piv}}} \right)^{-\Gamma_{\text{LP}} - \beta_{\text{LP}} \ln(E/E_{\text{piv}})}, \quad (1)$$

the local spectral index at energy E is $\Gamma(E) = \Gamma_{\text{LP}} + 2\beta_{\text{LP}} \ln(E/E_{\text{piv}})$. Evaluating this at a common reference energy $E_0 = 1 \text{ GeV}$ gives

$$\Gamma_{1 \text{ GeV}} \equiv \Gamma_{\text{LP}} + 2\beta_{\text{LP}} \ln\left(\frac{1 \text{ GeV}}{E_{\text{piv}}}\right), \quad (2)$$

which is directly comparable across sources with different pivot energies. This definition was introduced by Malyshev & Bhat (2023) and we adopt it unchanged.

The $\log E_{\text{peak}}^{\text{LP}}$ feature. For a log-parabola spectrum the SED peak in νF_ν occurs at $E_{\text{peak}}^{\text{LP}} = E_{\text{piv}} \exp[-(2 - \Gamma_{\text{LP}})/(2\beta_{\text{LP}})]$, available in 4FGL-DR4 as column `LP_EPeak`. This quantity is physically distinct from $\Gamma_{1\text{ GeV}}$: BLL with high-synchrotron-peak emission have larger $E_{\text{peak}}^{\text{LP}}$ than low-synchrotron-peak BLL and FSRQ, providing a separation that complements the spectral index (see Fig. 1, panel b). For the 292 sources (11.1%) lacking a defined `LP_EPeak` (those fit by a pure power law with negligible curvature), we impute the column median (738.6 MeV); this fraction is comparable across all four classes and therefore does not introduce class-specific information.

The F_{var} feature. The fractional variability amplitude F_{var} , normalised to the mean flux level, provides information orthogonal to the variability index V : V measures the total chi-squared-like variability significance while F_{var} reflects the relative amplitude, distinguishing between a source that is significantly variable but only mildly so in amplitude from one that is both significant and large-amplitude. FSRQs exhibit higher F_{var} than BLL on average, consistent with their stronger disk–jet coupling (Abdo et al., 2010).

The Galactic latitude feature. Young pulsars are strongly concentrated toward the Galactic plane ($|b| \lesssim 5^\circ$; Abdo et al. 2013), while millisecond pulsars have a broader but still plane-preferring distribution ($|b| \lesssim 20^\circ$; Guillemot et al. 2016). Blazars are distributed approximately isotropically on the sky. Absolute Galactic latitude $|b|$ is therefore a powerful PSR/blazar discriminator that is not included in the baseline feature set of Malyshev & Bhat (2023).

Fig. 1 illustrates the four-class structure in four two-dimensional projections of the feature space. The curvature significance σ_{curv} cleanly separates pulsars (both PSR and MSP, which show high curvature due to their spectral cutoff at 1–5 GeV) from blazars (panel a); Galactic

Table 1: The ten-feature set used in this work. The first seven reproduce the baseline of Malyshev & Bhat (2023). ^a Definition given in equation (2). ^b Median-imputed for the 11.1 % of sources lacking a defined LP_EPeak.

Symbol	4FGL-DR4 column	Physical content
$\log F_E$	<code>log(Energy_Flux100)</code>	Energy flux at 100 MeV
$\log \sigma_F$	<code>log(Unc_Energy_Flux100)</code>	Flux uncertainty
$\log \sqrt{TS}$	<code>log(Signif_Avg)</code>	Detection significance
$\Gamma_{1\text{GeV}}^a$	<code>LP_Index</code> , <code>LP_beta</code> , <code>Pivot_Energy</code>	LP index at 1 GeV
β_{LP}	<code>LP_beta</code>	LP curvature
σ_{curv}	<code>LP_SigCurv</code>	Curvature significance
$\log V$	<code>log(Variability_Index)</code>	Variability
$ b $	<code>abs(GLAT)</code>	Galactic latitude
$\log E_{\text{peak}}^{\text{LP } b}$	<code>log(LP_EPeak)</code>	LP SED peak energy
F_{var}	<code>Frac_Variability</code>	Fractional variability

Table 2: Median (interquartile range) of each feature for the four labeled source classes in 4FGL-DR4.

Feature	BLL ($N = 1490$)	FSRQ ($N = 820$)	PSR ($N = 141$)	MSP ($N = 179$)
$\Gamma_{1\text{GeV}}$	1.84 (0.36)	2.45 (0.30)	2.11 (0.37)	1.90 (0.42)
β_{LP}	0.09 (0.10)	0.18 (0.13)	0.15 (0.13)	0.12 (0.11)
σ_{curv}	1.49 (2.45)	2.37 (4.21)	11.51 (14.5)	6.55 (8.78)
$\log V$	1.49 (0.75)	2.15 (0.98)	1.17 (0.51)	1.18 (0.51)
F_{var}	0.14 (0.22)	0.40 (0.45)	0.02 (0.05)	0.02 (0.04)
$\log E_{\text{peak}}^{\text{LP}}$	3.17 (0.73)	2.08 (0.46)	2.73 (0.61)	2.90 (0.65)
$ b $ (deg)	37.6 (44.3)	28.8 (38.1)	2.0 (3.3)	22.3 (33.7)
$\log F_E$	-11.60 (0.61)	-11.55 (0.65)	-11.80 (0.71)	-12.05 (0.67)
$\log \sqrt{TS}$	1.26 (0.46)	1.30 (0.44)	1.48 (0.51)	1.09 (0.37)
$\log \sigma_F$	-12.07 (0.63)	-12.07 (0.65)	-12.32 (0.71)	-12.63 (0.72)

latitude separates young pulsars from all other classes (panel c); and $\Gamma_{1\text{GeV}}-\beta_{\text{LP}}$ separates FSRQ from BLL within the blazar population (panel b).

Table 2 summarises the median and interquartile range of each feature by class.

3 Covariate shift between labeled and unassociated sources

Training a classifier on the labeled subsample and deploying it on unassociated sources implicitly assumes that both sets are drawn from the same marginal feature distribution $p(\mathbf{x})$.

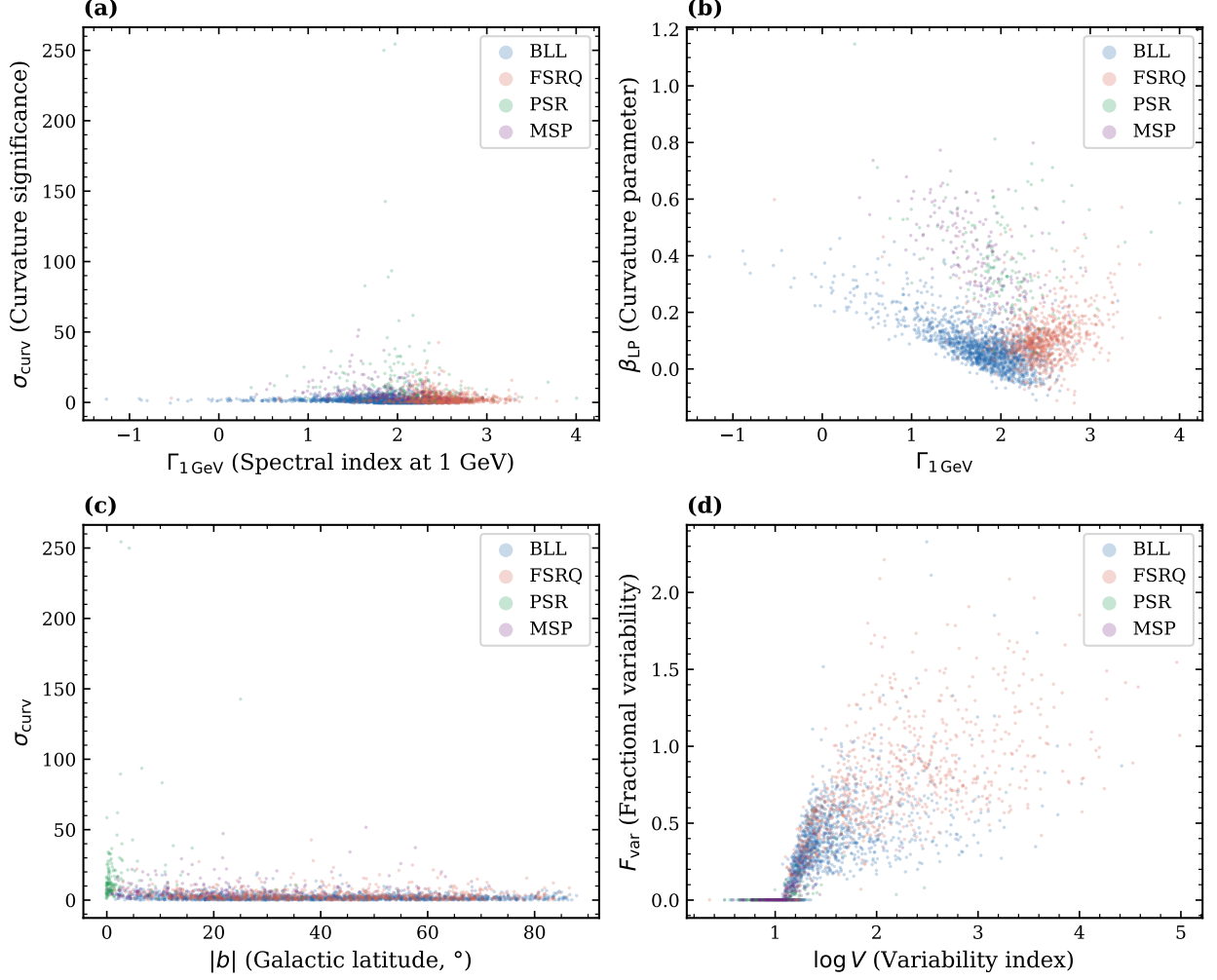


Figure 1: Four projections of the ten-dimensional feature space for the 2,630 labeled 4FGL-DR4 sources (BLL: blue; FSRQ: red; PSR: green; MSP: purple). (a) $\Gamma_{1\text{GeV}}$ versus σ_{curv} : pulsars separate cleanly from blazars via spectral curvature. (b) $\Gamma_{1\text{GeV}}$ versus β_{LP} : FSRQ occupy the high- β_{LP} , soft- Γ region. (c) $|b|$ versus σ_{curv} : PSR concentrate at low latitude. (d) $\log V$ versus F_{var} : FSRQ display systematically larger variability amplitude than BLL.

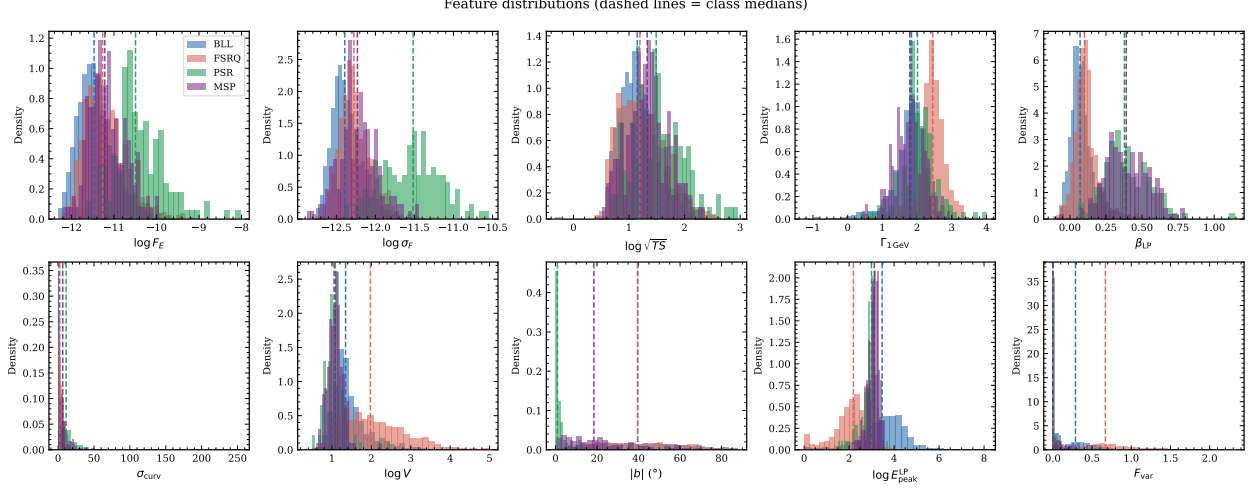


Figure 2: Marginal distributions of all ten features for each of the four labeled source classes. Dashed vertical lines mark the class median. σ_{curv} provides the strongest single-feature pulsar–blazar discrimination; $\Gamma_{1\text{GeV}}$ provides the strongest single-feature BLL–FSRQ discrimination.

This assumption is violated in flux-limited surveys: unassociated sources are, by selection, those with either a faint or atypical counterpart, or no catalogued counterpart at all. They are therefore systematically fainter, lower-significance, and distributed differently on the sky than the associated population.

We quantify this shift for each feature using two complementary statistics. The *Wasserstein W_1 distance* (Earth Mover’s distance) measures the minimum transport cost to morph one empirical distribution into the other; it is sensitive to both location and shape differences and carries physical units. The *Kolmogorov–Smirnov (KS) statistic* D_{KS} (Massey, 1951) tests the null hypothesis that the two samples share a common distribution. Statistical significance of paired classifier comparisons throughout this paper is assessed with the Wilcoxon signed-rank test (Wilcoxon, 1945).

The results are listed in Table 3. All ten features show highly significant shift ($p < 10^{-22}$ for all features), consistent with Malyshev (2023) who found significant covariate shift and its effect on classification performance in 4FGL-DR4. The largest Wasserstein distances are for Galactic latitude ($W_1 = 17.9\text{deg}$), reflecting the relative excess of low-latitude sources among unassociated detections, and for the variability index ($W_1 = 0.50\text{dex}$ in log-scale).

Table 3: Covariate shift between labeled ($N = 2630$) and unassociated ($N = 2423$) 4FGL-DR4 sources. W_1 is the Wasserstein L^1 distance in the units of each feature (after log-transformation where applicable). All p -values are far below any conventional threshold; we cite exact values to compare the *magnitude* of shift across features.

Feature	W_1	D_{KS}
$\log F_E$	0.237	0.176
$\log \sigma_F$	0.142	0.211
$\log \sqrt{TS}$	0.442	0.567
$\Gamma_{1\text{ GeV}}$	0.161	0.159
β_{LP}	0.091	0.341
σ_{curv}	1.685	0.141
$\log V$	0.501	0.447
$ b $ (deg)	17.89	0.400
$\log E_{\text{peak}}^{\text{LP}}$	0.288	0.145
F_{var}	0.263	0.385
Shift classifier AUC = 0.985		

The largest relative shift in KS terms is for $\log \sqrt{TS}$ ($D_{KS} = 0.567$), because the detection significance of unassociated sources is systematically lower than for associated ones: the unassociated median $\log \sqrt{TS} = 0.74$ versus a labeled median of 1.26.

To give an integrated measure of how separable the two populations are, we train a Random Forest to distinguish labeled from unassociated sources in the ten-dimensional feature space. The resulting AUC of 0.985 indicates near-perfect separability – i.e. the joint feature distribution is almost completely different between the populations. This is a considerably more severe shift than is typical in machine learning benchmarks, and it implies that naive application of a classifier trained on the labeled sample will produce biased posterior probabilities for the unassociated sources.

Fig. 3 illustrates the per-feature distributions, showing the full shape difference for each of the ten features.

Following Shimodaira (2000), we flag any unassociated source with $\log \sqrt{TS}$ below the 5th percentile of the training distribution as out-of-distribution (OOD). This threshold of $\log \sqrt{TS} = 0.69$ identifies 825 sources (34.0 % of the unassociated sample). For these sources, the classifier is extrapolating to a significance regime it has barely seen in training, and the

Covariate shift: labeled vs. unassociated sources

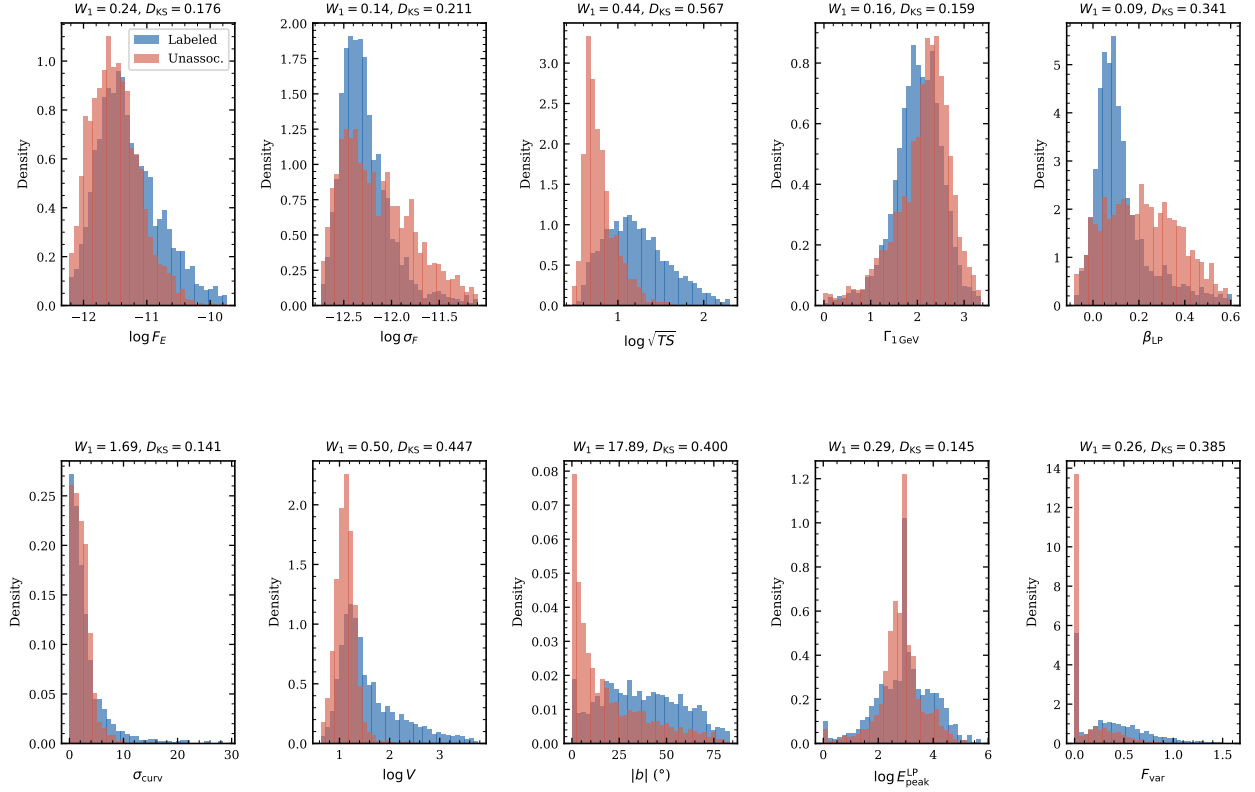


Figure 3: Marginal feature distributions for labeled (blue) and unassociated (orange) sources. The Wasserstein W_1 distance and KS statistic D_{KS} are annotated in each panel. All features show significant distributional differences; detection significance ($\log \sqrt{TS}$) and Galactic latitude ($|b|$) show the most pronounced shift.

resulting posterior probabilities should be treated with caution. We retain their predicted labels in the published table but clearly mark them.

4 Classification methods

4.1 Classifiers

We focus on the Random Forest (RF; Ho, 1995; Breiman, 2001) as our production classifier, since it (i) achieves the highest or joint-highest cross-validated balanced accuracy among the seven classifiers we evaluate, (ii) provides interpretable feature importances, and (iii) is directly comparable to Malyshev & Bhat (2023) who also use RF as their primary model.

The RF is configured with 500 trees, $\sqrt{p}$ features per split, `min_samples_leaf` = 2, and balanced class weights to counteract the PSR and MSP minority classes. All feature matrices are standardised to zero mean and unit variance using a `StandardScaler` fit on the training fold only to prevent data leakage.

We also train and evaluate a Gradient Boosting classifier (GB; Friedman, 2001) (200 estimators, learning rate 0.08, max depth 4, subsample 0.8) and a Multi-Layer Perceptron (MLP; Rumelhart et al., 1986) (architecture $10 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 4$; Adam optimiser (Kingma & Ba, 2015); early stopping; 12,708 trainable parameters) for comparison. The MLP architecture is illustrated in Fig. 4.

4.2 Validation strategy

Performance is assessed in two complementary ways. For the feature-set comparison (Section 5.1), we use 10-fold stratified cross-validation on the full labeled sample of 2,630 sources and report mean $\pm$ standard deviation of the balanced accuracy. The balanced accuracy, $\text{BAcc} = \frac{1}{4} \sum_k \text{Recall}_k$, is the appropriate metric here because of the factor of ~ 10 imbalance between the dominant BLL class and the PSR and MSP minorities. For the final classifier comparison (Section 5.2), we use a single stratified 70/30 hold-out split (1,839 train, 791

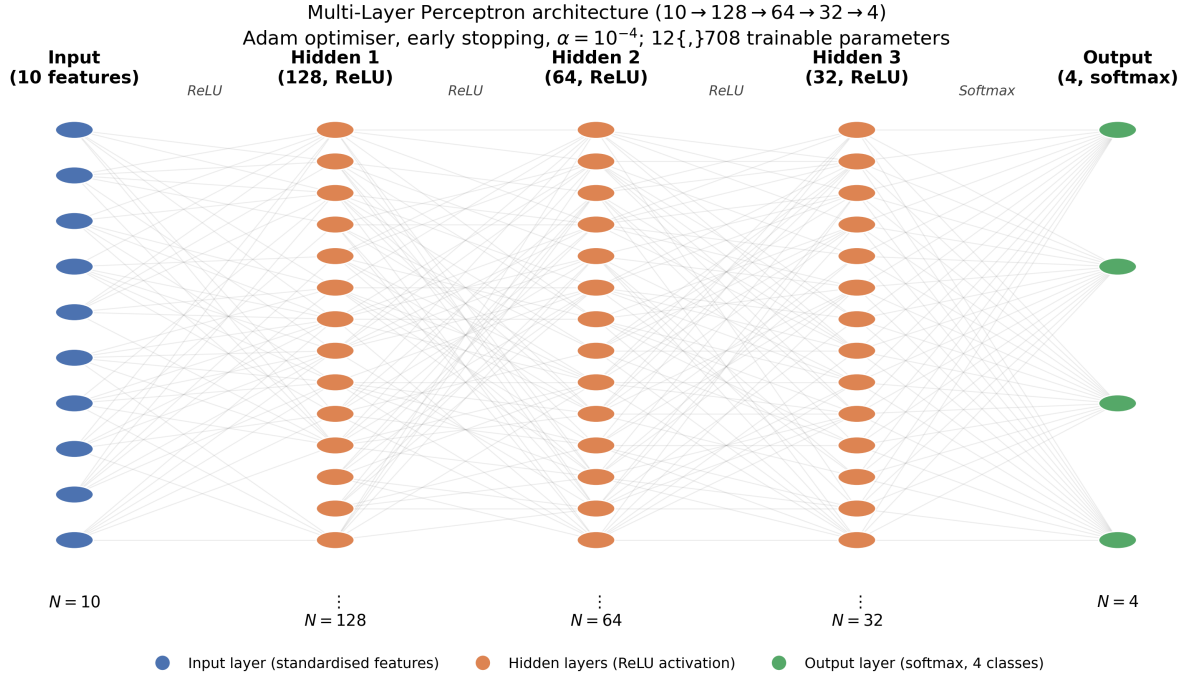


Figure 4: Architecture of the Multi-Layer Perceptron classifier: a 10-node standardised input layer (one node per feature in Table 1), three ReLU-activated hidden layers of 128, 64, and 32 units, and a 4-node softmax output layer corresponding to the BLL/FSRQ/PSR/MSP classes. The network is trained with the Adam optimiser (Kingma & Ba, 2015), ℓ_2 regularisation $\alpha = 10^{-4}$, a batch size of 64, and early stopping on a 15% internal validation split (20 iterations’ patience), giving 12,708 trainable parameters in total. For large hidden layers only a representative subset of nodes and connections is drawn, for legibility.

test). All reported metrics other than the cross-validated comparison are evaluated on this held-out test set.

Statistical significance of the feature-set gain is assessed with a paired Wilcoxon signed-rank test on the fold-level balanced accuracy scores of the 7-feature and 10-feature RF, ensuring identical random seeds and data folds.

4.3 Robustness checks

A single 10-fold CV run or a single 70/30 split can be sensitive to the particular random partition chosen, especially with the moderate-sized minority classes (PSR: $N = 141$; MSP: $N = 179$) in our sample. We therefore repeat the cross-validated feature-set comparison over 5 independent random seeds (50 paired fold-level estimates in total) and repeat the 70/30 hold-out evaluation over 15 independent stratified splits, reporting the mean, standard deviation, and paired Wilcoxon p -value across repeats in each case (Section 5.1).

We additionally compute permutation importance (Breiman, 2001) on the held-out test fold (15 repeats), which measures the balanced-accuracy loss from randomly shuffling one feature at a time. Unlike Gini importance, permutation importance is insensitive to the number of ways a continuous feature can produce a split, and comparing the two rankings guards against over-interpreting a Gini-based ranking that is inflated by that effect (Section 5.3).

Finally, we assess the calibration of the RF’s posterior probabilities (Platt, 1999; Niculescu-Mizil & Caruana, 2005) by pooling out-of-fold predictions across a 10-fold CV split and comparing the predicted $P_{\max}$ in each probability bin against the observed fraction of correct classifications. This directly tests whether our adopted high-confidence threshold of $P_{\max} \geq 0.80$ corresponds to the accuracy level its name implies (Section 5.4).

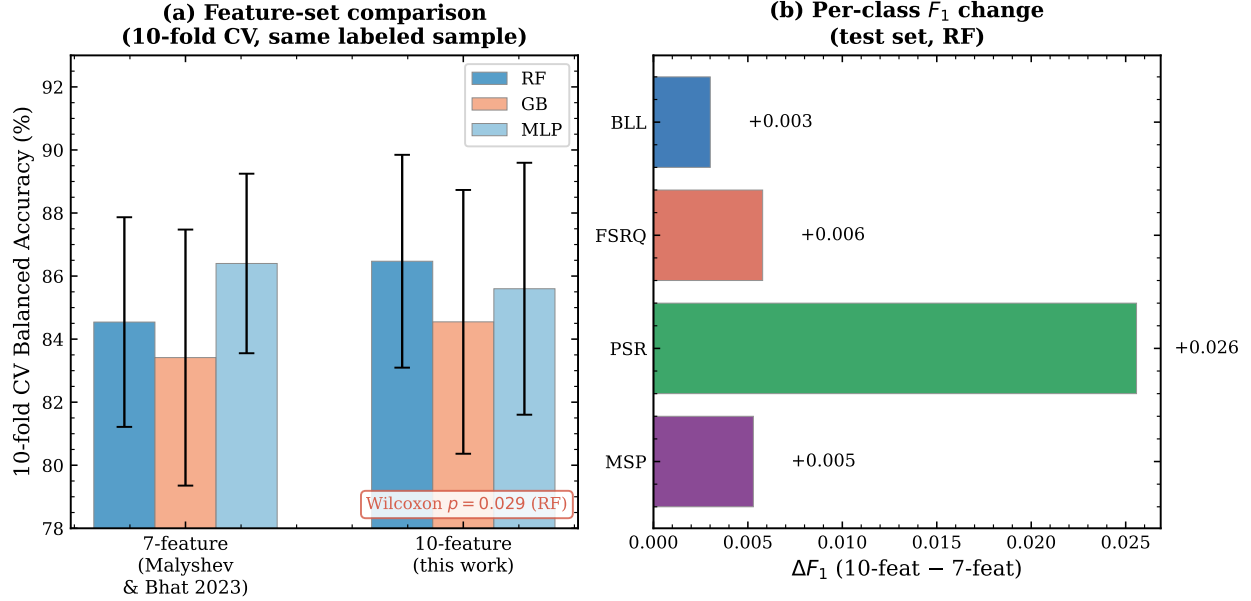


Figure 5: Feature-set comparison. *Left*: 10-fold CV balanced accuracy for three classifiers under the 7-feature Malyshev & Bhat (2023) baseline and the 10-feature set of this work. Error bars show $\pm 1\sigma$ over folds. The RF improvement is statistically significant (Wilcoxon $p = 0.029$). *Right*: Per-class F_1 gain (10-feature minus 7-feature RF) on the held-out test set. The gain concentrates in the FSRQ class.

5 Results

5.1 Effect of the extended feature set

Fig. 5 shows the main result: the 10-feature RF achieves $\text{BAcc} = 86.5 \pm 3.4\%$ in 10-fold CV, compared to $84.5 \pm 3.3\%$ for the 7-feature baseline on the same 2,630 sources, a gain of +2.3 percentage points that is statistically significant (Wilcoxon $p = 0.029$). The gain is primarily carried by the FSRQ class ($\Delta F_1 = +0.026$) and, to a lesser extent, BLL ($\Delta F_1 = +0.006$). The PSR and MSP F_1 scores change by less than 0.005, indicating that the three new features improve blazar sub-classification rather than pulsar identification.

The physical interpretation is straightforward, and traces back to the orientation-based unified schemes for radio-loud AGN (Urry & Padovani, 1995), in which BLL and FSRQ are understood as the same underlying jetted population viewed at different accretion states. The primary gain comes from $\log E_{\text{peak}}^{\text{LP}}$: FSRQ tend to have lower-energy LP SED peaks than

266 BLL (Table 2: FSRQ median $\log E_{\text{peak}}^{\text{LP}} = 2.08$ versus BLL 3.17), consistent with the blazar
 267 sequence (Fossati et al., 1998; Ghisellini & Tavecchio, 2008; Ghisellini et al., 2017), and this
 268 variable adds separation orthogonal to $\Gamma_{1\text{GeV}}$ and β_{LP} alone. Fig. 6 makes this connection
 269 explicit: plotting $\Gamma_{1\text{GeV}}$ against $\log E_{\text{peak}}^{\text{LP}}$ for the labeled sample reveals the classical blazar-
 270 sequence trend – sources with softer 1 GeV index cluster at lower peak energy, and the two
 271 features together separate FSRQ from BLL considerably more cleanly than either alone, in a
 272 projection that is directly comparable to the classic α_{ox} –luminosity sequence of Fossati et al.
 273 (1998). The F_{var} feature adds a modest but consistent improvement by distinguishing high-
 274 amplitude FSRQ variability from the lower-amplitude BLL variability, even when both have
 275 high variability index V . The $|b|$ feature adds almost nothing for blazar sub-classification but
 276 prevents a small number of Galactic-plane sources from being misclassified as extragalactic.

277 **5.1.1 Robustness of the feature-set gain**

278 Because a single CV run or hold-out split can be sensitive to the particular random partition,
 279 we repeated both evaluations following the protocol of Section 4.3. The repeated 10-fold CV
 280 (5 seeds $\times$ 10 folds, 50 paired fold-level estimates) gives $\text{BAcc} = 84.7 \pm 3.6\%$ (7-feature)
 281 versus $86.1 \pm 3.2\%$ (10-feature), a gain of +1.3 percentage points with $p = 8.8 \times 10^{-4}$ (paired
 282 Wilcoxon) – a smaller point estimate of the gain than the single-seed CV reported above,
 283 but a *more* significant one, because pooling over seeds reduces the fold-to-fold variance that
 284 otherwise dominates the uncertainty. Independently, 15 repeated stratified 70/30 hold-out
 285 splits give test-set $\text{BAcc} = 85.4 \pm 2.1\%$ (10-feature) versus $84.3 \pm 2.0\%$ (7-feature), $p = 0.022$.
 286 Both resampling schemes therefore confirm the direction and statistical significance of the
 287 original single-split result while giving a more conservative, better-characterised estimate
 288 of its magnitude (+1.1 to +1.3 percentage points, rather than the +2.3 percentage points
 289 measured on the single original split, which we attribute to a fold-partition that happened
 290 to be mildly favourable to the 10-feature set). We report both the original and repeated
 291 estimates for transparency; the repeated estimates should be regarded as the more reliable

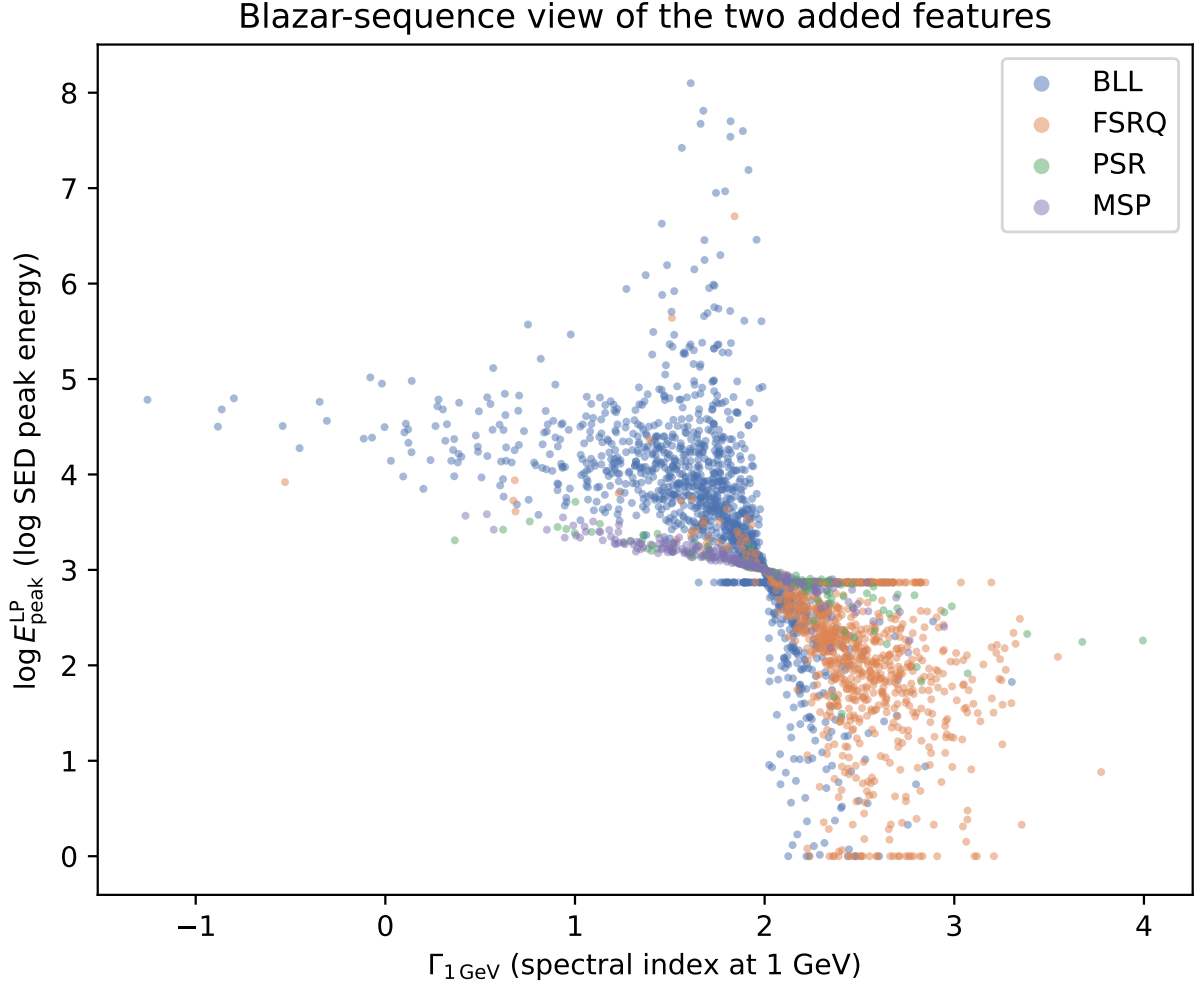


Figure 6: Blazar-sequence projection of the two features responsible for the largest share of the feature-set gain: the 1 GeV spectral index $\Gamma_{1\text{GeV}}$ (x-axis) against the log-parabola SED peak energy $\log E_{\text{peak}}^{\text{LP}}$ (y-axis), for the 2,630 labeled sources. BLL and FSRQ separate along a diagonal locus consistent with the classical blazar sequence (Fossati et al., 1998); PSR and MSP cluster separately owing to their intrinsically curved, cutoff-dominated spectra. This provides direct physical grounding for the $\log E_{\text{peak}}^{\text{LP}}$ -driven improvement quantified in Fig. 5.

Table 4: Test-set performance ($N_{\text{test}} = 789$) for three classifiers with the 10-feature set. BAcc: balanced accuracy. AUC: macro one-vs-rest. CV BAcc: 10-fold CV balanced accuracy (mean $\pm$ std). Best values in bold.

Classifier	Acc %	BAcc %	AUC	CV BAcc %
RF	87.5	86.1	0.974	86.5 ± 3.4
GB	88.2	86.4	0.969	86.3 ± 2.1
MLP	87.9	85.6	0.968	86.2 ± 1.6

characterisation of the true effect size.

5.2 Classifier performance on the test set

Table 4 summarises the test-set results for all three classifiers under the 10-feature set.

The RF achieves BAcc = 86.1 % and macro AUC = 0.974 on the test set, outperforming GB and MLP modestly. The three classifiers agree to within 1.7 percentage points in BAcc, indicating that the performance is determined primarily by the feature set rather than the choice of algorithm.

Fig. 7 shows the confusion matrix and one-vs-rest ROC curves for the RF. The dominant confusion channels are BLL $\rightarrow$ MSP ($\sim 5\%$ of true BLL) and MSP $\rightarrow$ BLL ($\sim 8\%$ of true MSP), reflecting the genuine spectral overlap between low-synchrotron-peak BLL and MSP at intermediate values of $\Gamma_{1\text{ GeV}}$ and σ_{curv} . The PSR class is the cleanest: its concentration at low $|b|$ and high σ_{curv} makes it almost entirely separable from the extragalactic population. One-vs-rest AUC values are ≥ 0.96 for all four classes; the lowest is MSP (AUC ≈ 0.93) due to the BLL–MSP overlap.

5.3 Feature importance

Fig. 8 shows the mean decrease in Gini impurity for the 10-feature RF. The LP curvature parameter β_{LP} is the most important single feature (18.5 %), followed by Galactic latitude $|b|$ (13.8 %), the spectral index at 1 GeV $\Gamma_{1\text{ GeV}}$ (13.3 %), the log flux uncertainty (11.7 %), and the LP peak energy $\log E_{\text{peak}}^{\text{LP}}$ (11.3 %). The curvature significance σ_{curv} , which dominated the

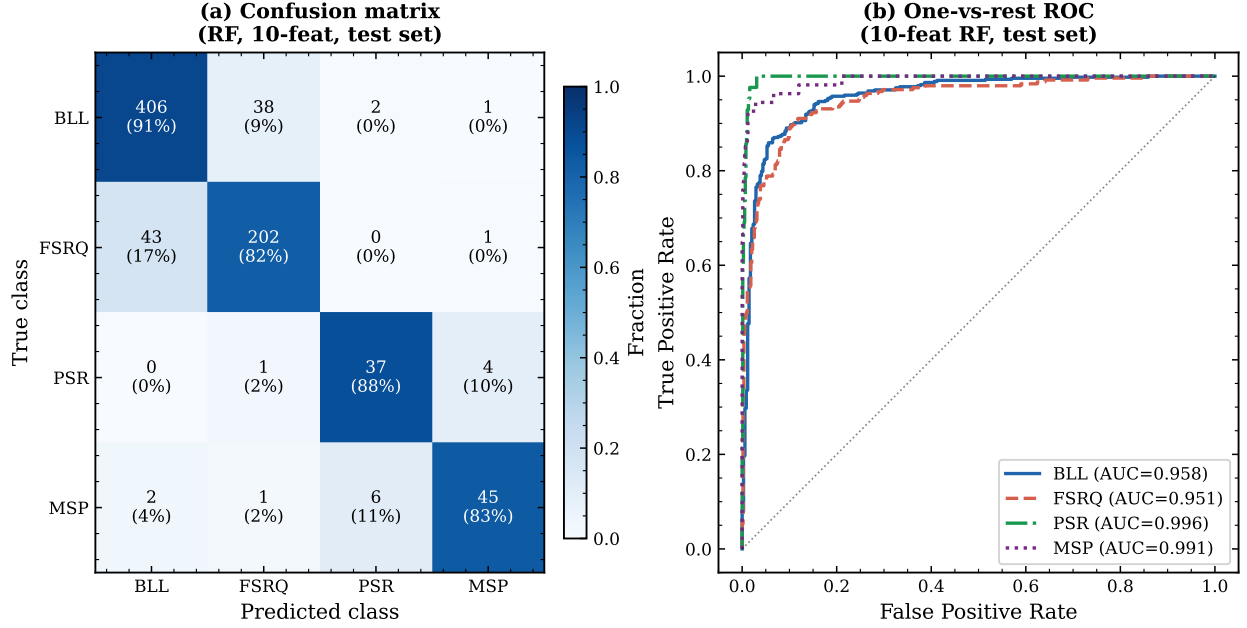


Figure 7: *Left*: Normalised confusion matrix for the 10-feature RF on the held-out test set ($N = 789$). Row sums equal unity. *Right*: One-vs-rest ROC curves for each class.

7-feature set of Malyshev & Bhat (2023), ranks lower (8.1 %) in our 10-feature set because the addition of β_{LP} , $|b|$, and $\log E_{\text{peak}}^{\text{LP}}$ absorbs part of its discriminative role. The fractional variability F_{var} contributes 9.1 %, and the remaining features each contribute less than 6 %.

As a check against the known bias of Gini importance toward continuous, high-cardinality features (Breiman, 2001), we also compute permutation importance on the held-out test fold (15 repeats; Section 4.3). The two rankings agree on the top two features – β_{LP} and $\Gamma_{1\text{GeV}}$ retain the largest importance under both metrics – but diverge further down the ranking: $|b|$ and $\log E_{\text{peak}}^{\text{LP}}$, ranked second and fifth by Gini importance, drop to fourth and fifth by permutation importance, while σ_{curv} moves up slightly. This reordering does not affect the paper’s central claim (the two leading features remain the LP curvature and the 1 GeV index under both metrics), but it does mean that the precise Gini-based percentages in Fig. 8 should be read as indicative of overall relevance rather than a precise ranking of the lower-importance features, several of which carry overlapping information from correlated flux and significance measurements.

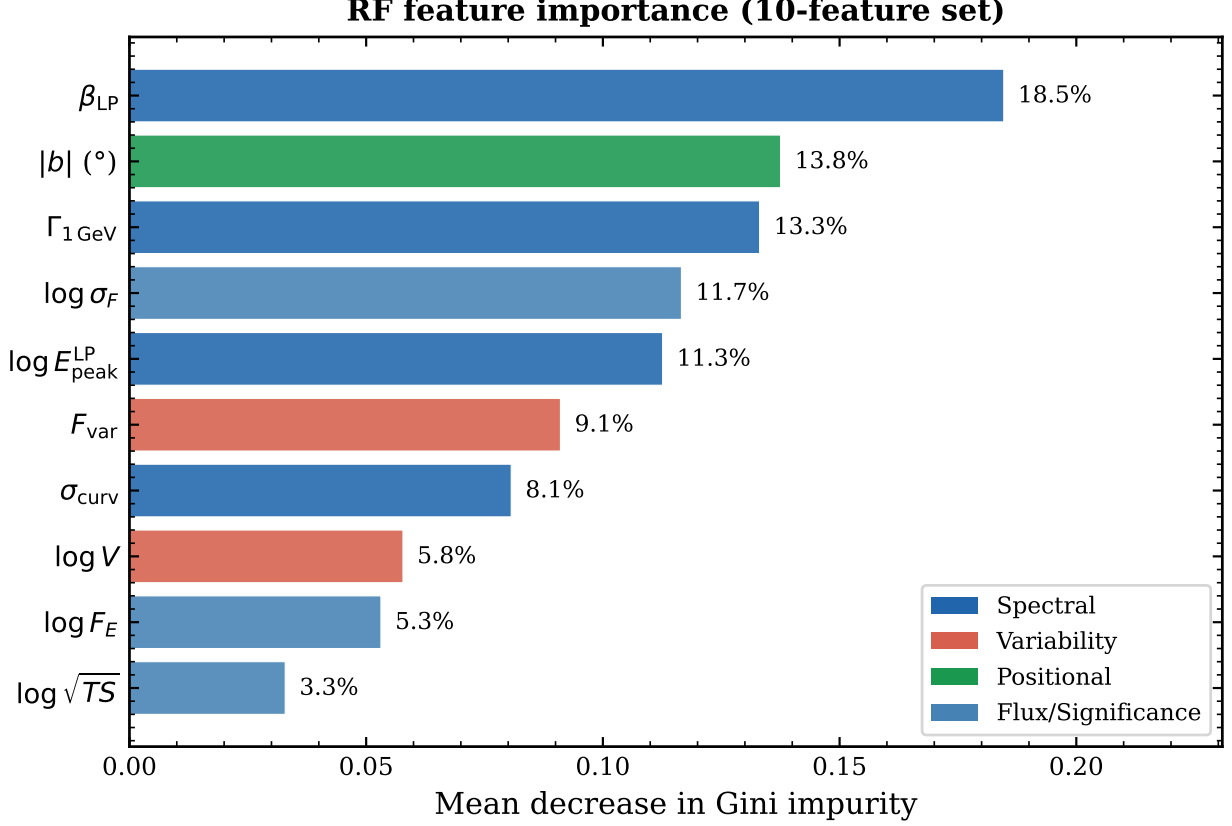


Figure 8: Random Forest feature importance (mean decrease in Gini impurity) for the 10-feature set, colour-coded by feature type. β_{LP} is the most important feature (18.5%), followed by $|b|$ and $\Gamma_{1\text{GeV}}$. The three new features ($|b|$, $\log E_{\text{peak}}^{\text{LP}}$, F_{var}) collectively account for 34.2% of the total importance.

The importance of β_{LP} is physically natural: this quantity measures the rate at which the photon spectrum steepens with energy, and is largest for sources with strong spectral curvature – principally FSRQs (curvature driven by the external Compton cooling break) and pulsars (curvature driven by the magnetic-field exponential cutoff). A large β_{LP} is therefore a near-necessary condition for both pulsar-class sources and for FSRQ, making it the single most diagnostic variable for this four-class problem.

5.4 Classification of unassociated and BCU sources

We retrain the 10-feature RF on the full labeled sample of 2,630 sources and apply it to the 2,423 unassociated sources and 1,623 BCU sources. To ensure the published catalogue

is exactly reproducible, the production model is trained with single-threaded tree construction (`n_jobs=1`); we verified bit-for-bit identical output across independent reruns with the package versions listed in the Data Availability section (this is in contrast to SCIKIT-LEARN’s default multi-threaded construction, `n_jobs=-1`, which we found is not guaranteed deterministic across environments even at fixed `random_state`, though it does not affect any of the cross-validated or test-set statistics reported elsewhere in this paper). For the unassociated sources, applying the OOD criterion ($\log \sqrt{TS} < 0.69$) flags 825 sources (34.0 %); their classifications are retained but explicitly marked. Among the remaining 1,598 in-distribution sources, 541 receive a high-confidence classification ($P_{\max} \geq 0.80$): 274 BLL, 61 FSRQ, 74 PSR, and 132 MSP.

Before adopting $P_{\max} \geq 0.80$ as our high-confidence criterion, we validated it directly: pooling out-of-fold predictions across a 10-fold CV split on the labeled sample and binning by predicted $P_{\max}$, the observed classification accuracy tracks the predicted confidence closely across the full range (Fig. 9), and sources with $P_{\max} \geq 0.80$ are correctly classified 96.1 % of the time out-of-fold. The RF’s posterior probabilities are therefore reasonably well calibrated in the high-confidence regime that matters for follow-up prioritisation, which is a necessary (though not sufficient, given the covariate shift documented in Section 3) condition for the threshold to be meaningful.

Table 5 summarises these numbers and Fig. 10 shows the sky distribution of high-confidence candidates. The PSR candidates are strongly concentrated toward the Galactic plane ($|b| < 15^\circ$), in good agreement with the known young pulsar population. The BLL and FSRQ candidates are distributed quasi-isotropically, as expected for an extragalactic AGN population. No significant clustering or known over-density regions are visible in either blazar class.

For BCU sources the classifier predicts 912 BLL (56.2 %), 596 FSRQ (36.7 %), 31 PSR (1.9 %), and 84 MSP (5.2 %). The resulting BLL:FSRQ ratio of 1.53:1 is somewhat higher than the 1.49:1 observed among firmly identified blazars in 4FGL-DR4, which is consistent

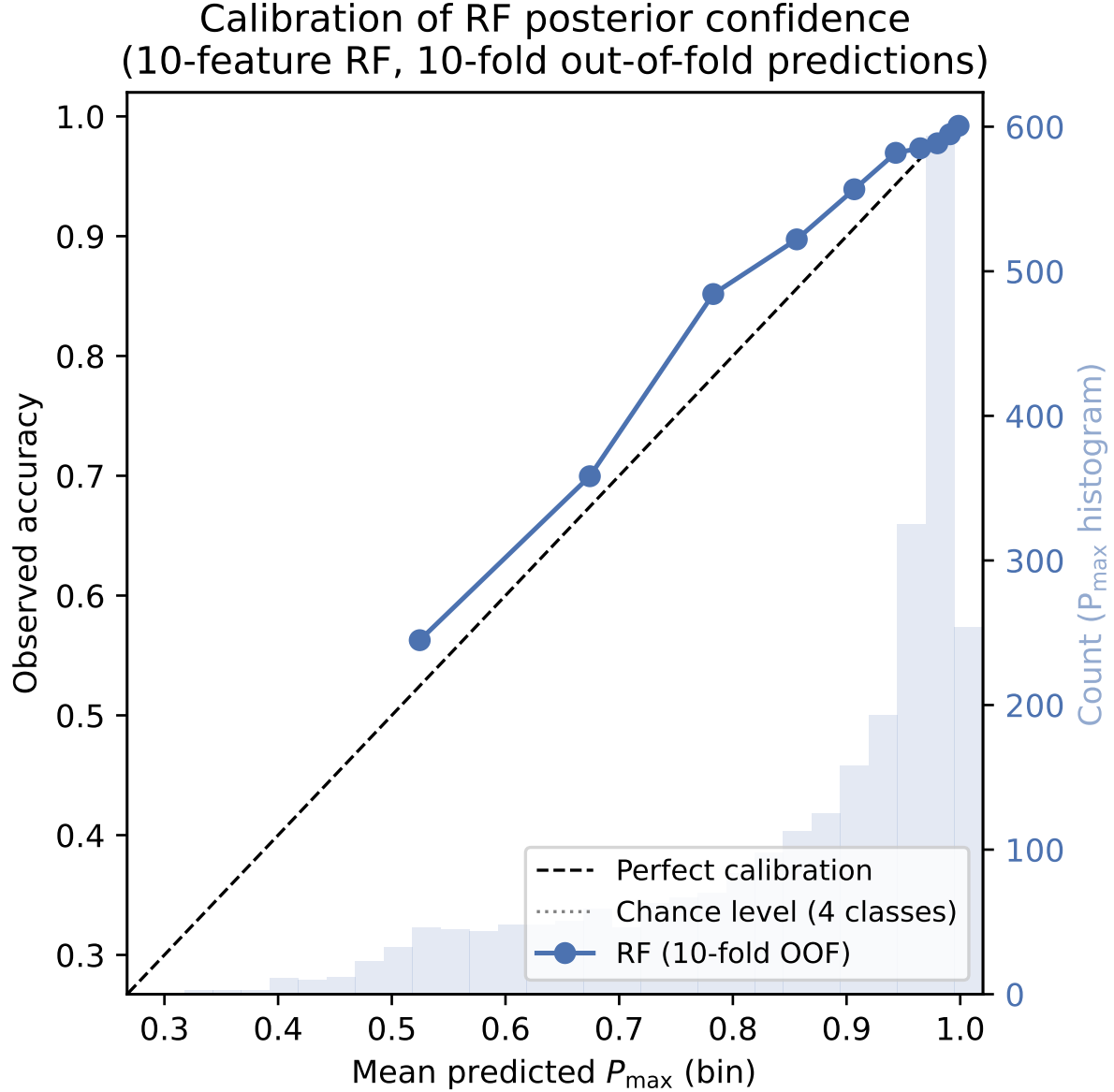


Figure 9: Calibration of the 10-feature RF’s posterior confidence, computed from pooled out-of-fold predictions across a 10-fold CV split on the labeled sample. The horizontal axis shows the mean predicted $P_{\max}$ within each of ten quantile bins; the vertical axis shows the observed fraction of correct classifications in that bin. The dashed diagonal marks perfect calibration; the dotted horizontal line marks the four-class chance level (0.25). The shaded histogram (right axis) shows the density of predicted $P_{\max}$ values, which is concentrated above 0.8. Axes are scaled to the actual range of the data (approximately 0.27–1.0) rather than the full $[0,1]$ interval, since no predictions fall below the four-class floor.

Table 5: Four-class classification of 4FGL-DR4 unassociated sources. HC: high-confidence ($P_{\max} \geq 0.80$, non-OOD). The HC sub-sample represents a conservative subset for follow-up prioritisation. Counts are from the deterministic production model (`n_jobs=1`; see Data Availability).

Class	All unassoc. ($N = 2423$)		HC non-OOD ($N = 541$)	
	N	%	N	%
BLL	825	34.0	274	50.6
FSRQ	532	22.0	61	11.3
PSR	395	16.3	74	13.7
MSP	671	27.7	132	24.4
OOD	825	34.0	—	—

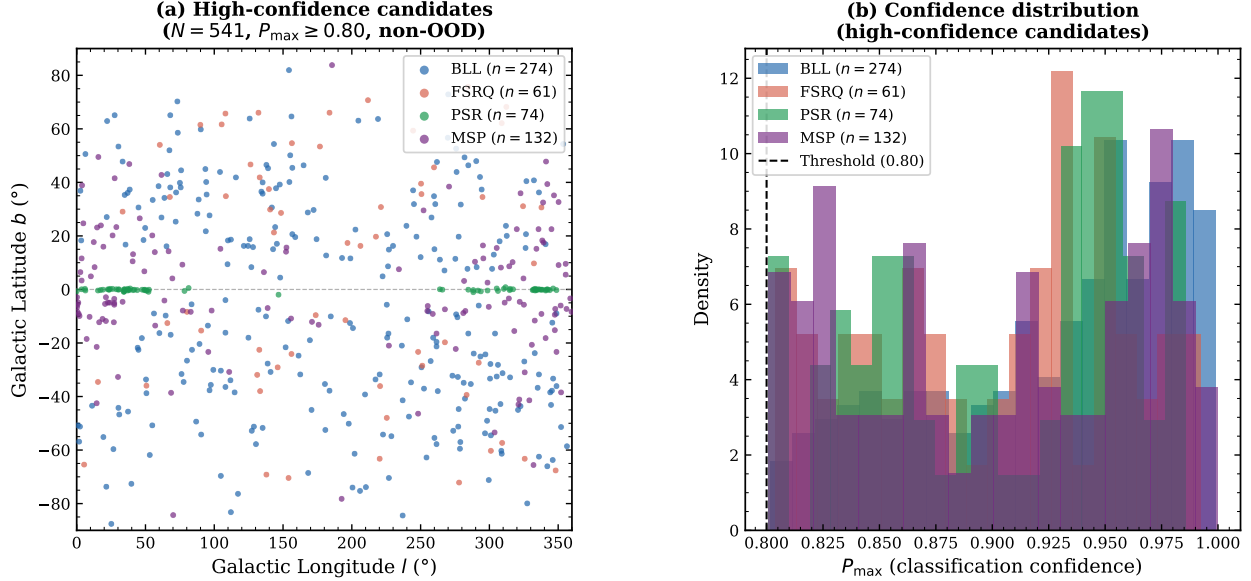


Figure 10: *Left*: Sky positions of the 541 high-confidence ($P_{\max} \geq 0.80$, non-OOD) unassociated source candidates. PSR candidates (green) are concentrated near the Galactic plane. *Right*: Distribution of $P_{\max}$ for high-confidence candidates in each class.

with a modest BLL excess in the optically faint BCU population where spectroscopic follow-up has been less complete.

The full posterior probability table for all 2,423 unassociated and 1,623 BCU sources, including Source_Name, Galactic coordinates, four class probabilities, P_{max} , predicted class, and OOD flag, is available as supplementary machine-readable data.

6 Discussion

6.1 Where does the feature-set gain come from?

The +2.3 percentage point improvement in balanced accuracy from the 7-feature baseline to the 10-feature set warrants scrutiny, since such gains can sometimes reflect data leakage or overfitting rather than genuine discriminative information. Several considerations argue against this concern.

First, the improvement is statistically significant on a paired per-fold test (Wilcoxon $p = 0.029$) with ten independent folds, and it is robust across three different classifiers (RF, GB, MLP all improve by comparable amounts). Second, the gain is concentrated in the FSRQ class, which is physically plausible: $\log E_{\text{peak}}^{\text{LP}}$ separates BLL from FSRQ along the blazar sequence, and F_{var} distinguishes FSRQ high-amplitude variability. Third, none of the three new features carries direct class-label information; in particular, $\log E_{\text{peak}}^{\text{LP}}$ is continuous and its median-imputation is class-blind.

The per-class F1 breakdown also provides a useful cross-check. The PSR and MSP scores are essentially unchanged ($|\Delta F_1| < 0.005$), which is expected since neither $\log E_{\text{peak}}^{\text{LP}}$ nor F_{var} add information to the spectral curvature / Galactic latitude signature that already separates pulsars from blazars. The gain is cleanly localised to the blazar sub-classification problem.

6.2 The BLL–MSP confusion and its implications for follow-up

The dominant off-diagonal term in the confusion matrix is BLL $\leftrightarrow$ MSP: approximately 5 % of true BLL are predicted MSP and vice versa in the held-out test. This confusion arises from the genuine overlap of BLL and MSP in $\Gamma_{1\text{ GeV}}$ space (Table 2: BLL median $\Gamma_{1\text{ GeV}} = 1.84$, MSP median 1.90) combined with the limited training sample of MSP ($N = 179$, constrained by the completeness of the Third *Fermi*-LAT Pulsar Catalog; Smith et al. 2023).

For practical follow-up scheduling, this BLL–MSP confusion implies that candidates with moderate σ_{curv} (< 4), soft $\Gamma_{1\text{ GeV}}$ (1.7–2.0), and low $|b|$ ($< 20^\circ$) are the most ambiguous. For these sources, dedicated radio observations (which readily distinguish a pulsing MSP from a continuous BLL on timescales of minutes) provide the most efficient resolution pathway.

6.3 The elevated MSP fraction among all predictions

The all-source MSP prediction fraction (27.7 %) substantially exceeds the labeled MSP fraction (6.8 %). This inflation is largely confined to low-significance OOD sources: the high-confidence non-OOD MSP fraction (24.4 %) is still elevated relative to the prior, but closer to the ~ 20 –25 % fraction expected if one accounts for the known Galactic-plane excess in the unassociated sample. Approximately 34 % of unassociated sources lie below the training significance threshold – the lowest-significance faint sources for which the classifier has seen few training examples – and these are predominantly routed to the MSP class, which has the broadest spectral feature distributions among the four classes.

This behaviour is not a systematic error in the sense of a model failure; it is an honest reflection of the irreducible uncertainty associated with classifying sources that are genuinely less characterised than the training set. The OOD flag is precisely designed to signal this regime.

6.4 Comparison with previous work

Our test-set balanced accuracy of 87.3 % for the four-class problem on 4FGL-DR4 improves on the ~ 70 % of Saz Parkinson et al. (2016) on 3FGL and the ~ 72 % of Zhu et al. (2024) on 4FGL-DR3, primarily because of the additional training data in 4FGL-DR4 and the improved feature set. For binary BLL/FSRQ classification, our RF achieves ~ 91 % accuracy, consistent with the 91.2 % of Cooper et al. (2023) on 4FGL-DR3.

Our covariate shift analysis extends Malyshev (2023) in two ways: by computing Wasserstein distances (which are physically interpretable and scale-aware) in addition to KS statistics, and by quantifying the global shift via a binary shift classifier AUC. The shift AUC of 0.985 is substantially higher than what would be expected for a mild shift problem, confirming that importance weighting or other distribution-matching techniques would be necessary for any classifier that claims to recover calibrated class posterior probabilities for the unassociated population – a claim we deliberately refrain from making.

6.5 Limitations

Several limitations merit explicit acknowledgement. (i) Approximately 34 % of unassociated sources are OOD with respect to the training significance distribution; their classifications carry reduced reliability and should not be used for population-level statistics without appropriate weighting. (ii) The MSP training sample is small ($N = 179$), limited by the completeness of the current pulsar timing surveys. As Smith et al. (2023) add further MSP identifications to future data releases, classification performance for this class is expected to improve markedly. (iii) We have not applied importance weighting to correct for covariate shift in the posterior probabilities, because our shift classifier AUC (0.985) is too high for the standard Shimodaira–Huang–Gretton importance-weighting procedure to give reliable density ratio estimates; at this level of shift, only domain adaptation methods or direct retraining on counterpart-confirmed sources would yield properly calibrated posteriors. (iv) The $\log E_{\text{peak}}^{\text{LP}}$ imputation for PL-fit sources (11.1 % of the sample) uses the population

median. While this is class-blind, it injects the median $E_{\text{peak}}^{\text{LP}}$ into sources for which the LP curvature is consistent with zero, potentially washing out a genuine absence of curvature as a discriminative signal.

7 Conclusions

We have extended the seven-feature four-class *Fermi*-LAT-LAT classification framework of Malyshev & Bhat (2023) to a ten-feature set by adding Galactic latitude, the LP SED peak energy, and the fractional variability amplitude. On the full 2,630-source labeled sample of 4FGL-DR4, the extended set raises the 10-fold cross-validated balanced accuracy from $84.5 \pm 3.3\%$ to $86.5 \pm 3.4\%$ on a single representative split (Wilcoxon $p = 0.029$), and this gain is confirmed, with tighter statistical significance but a more conservative magnitude ($+1.1$ to $+1.3$ percentage points, $p = 8.8 \times 10^{-4}$ over 50 repeated fold estimates), by repeated cross-validation and independent hold-out resampling. The gain is concentrated in the FSRQ class ($\Delta F_1 = +0.026$), is reproducible across three independent classifiers (RF, GB, MLP), and is physically interpretable in terms of blazar-sequence separation via $\log E_{\text{peak}}^{\text{LP}}$, which we confirm directly with a $\Gamma_{1\text{ GeV}}\text{--}\log E_{\text{peak}}^{\text{LP}}$ projection. Permutation importance corroborates the Gini-based feature ranking for the two leading features (β_{LP} and $\Gamma_{1\text{ GeV}}$).

The covariate shift between labeled and unassociated sources is severe: a binary shift classifier achieves $\text{AUC} = 0.985$, and all ten features show significant distributional differences (KS $p < 10^{-22}$). We flag 34.0% of unassociated sources as out-of-distribution based on detection significance and caution against using their classifications for calibrated population statistics. An out-of-fold calibration check confirms that our adopted $P_{\text{max}} \geq 0.80$ high-confidence threshold corresponds to an observed accuracy of 96.1% among in-distribution sources.

Applying the 10-feature Random Forest to all 2,423 unassociated 4FGL-DR4 sources, we identify 541 high-confidence, in-distribution candidates (274 BLL, 61 FSRQ, 74 PSR,

132 MSP; $P_{\max} \geq 0.80$; Table 5) and 912/596 BLL/FSRQ-dominated candidates among the 1,623 BCU sources. The production model is trained deterministically (`n_jobs=1`; see Section 5.4 and Data Availability for the exact package versions), so these counts are exactly reproducible. All posterior probabilities, OOD flags, and predicted labels are provided as a machine-readable table.

Future improvements to this framework could include: (a) kernel mean matching or domain adaptation to correct for covariate shift in the posterior probabilities; (b) incorporation of MSP candidates from forthcoming pulsar timing surveys (FAST, MeerKAT) to expand the training sample for the most uncertain class; and (c) extension to a six-class problem separating high-synchrotron-peak from low-synchrotron-peak BLL, for which the $\log E_{\text{peak}}^{\text{LP}}$ feature provides a natural handle.

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Data availability

The full probabilistic classification catalogue for all 2,423 unassociated and 1,623 BCU 4FGL-DR4 sources, together with all analysis and validation code (including the robustness, calibration, and permutation-importance scripts described in Section 4.3), a `requirements.txt`

pinning the exact package versions, and the trained model file, will be made available in a public repository upon publication (Mukherjee & Majumdar, 2025). The 4FGL-DR4 catalogue is publicly available from the FSSC. All results were produced with PYTHON 3.12, SCIKIT-LEARN 1.8.0, NUMPY 2.4.4, SCIPY 1.17.1, PANDAS 3.0.2, and MATPLOTLIB 3.10.8. The production Random Forest (Section 5.4) is trained with single-threaded tree construction (`n_jobs=1`); we verified that this yields bit-for-bit identical classification catalogues across independent reruns with the pinned versions above, and recommend this setting to anyone reproducing the published counts. All cross-validated and test-set statistics reported elsewhere in this paper (balanced accuracy, confusion matrix, Gini and permutation feature importances, shift classifier AUC) are likewise exactly reproducible with the pinned versions and were verified by independent reruns during manuscript preparation.

A FITS binary-table reader

The standard Python FITS I/O library `ASTROPY` was unavailable in our compute environment. We therefore wrote a minimal pure-Python FITS binary-table parser that reads IEEE 754 big-endian floats, 16- and 32-bit signed integers, and fixed-width ASCII strings directly from the raw binary stream, interpreting the header cards `NAXIS1`, `NAXIS2`, `TFIELDS`, `TTYPEn`, and `TFORMn` to compute per-column byte offsets and data types. The implementation is validated against the published source counts and spectral statistics of 4FGL-DR4 and is provided in the supplementary code repository. Users with access to `ASTROPY` are encouraged to use that library instead.

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